Name : Vinit Sandeep Karjekar

Designation : Data Analyst Intern

Joining Date : 01 / Feb / 2026

Product Line Profitability & Margin Performance Analysis for Nassau Candy Distributor

Executive Summary

This study presents a comprehensive analysis of product-level profitability and margin performance for Nassau Candy Distributor using sales, cost, and operational data.

The objective of this analysis is to identify high-performing products, detect margin inefficiencies, and evaluate division-level financial performance. The study applies statistical and exploratory data analysis techniques to uncover key drivers of profitability and operational risks.

The study provides actionable recommendations including pricing optimization, cost reduction strategies, and product portfolio rationalization to enhance profitability and operational efficiency.

Problem Statement

Currently, the organization lacks visibility into:

- Which product lines deliver the highest gross margin
- Whether high-sales products are actually profitable
- How profitability varies across product divisions
- Which products represent margin risk

Background and Context

For distributors like Nassau Candy, sales volume alone can be misleading when evaluating product performance. High sales do not necessarily translate into strong profitability.

Some products :

- Sell in high volume but generate low profit
- Consume disproportionate manufacturing or operational cost
- Appear successful but negatively impact overall margins

To remain competitive, the company must develop a data-driven understanding of:

- Which products truly drive profit
- Which divisions underperform financially
- Where pricing, sourcing, or product rationalization is required

Dataset Description

The dataset contains transactional and operational data including:

- Order details (Order ID, Order Date, Ship Date)
- Customer information (Region, City, State)
- Product attributes (Product Name, Division)
- Financial metrics (Sales, Cost, Gross Profit)
- Operational metrics (Units, Ship Mode)
- Additionally, factory-level data is used to map product manufacturing locations, enabling future logistics analysis.

Analytical Methodology

Data Cleaning & Validation

The dataset was preprocessed to ensure data accuracy:

- Validated cost and sales values
- Removed zero-sales and invalid records
- Handled missing unit values
- Standardized product and division labels

Analytical Methodology (Executive Summary)

Profitability Metrics

- Calculated Gross Margin (%), Profit per Unit, Revenue Contribution, and Profit Contribution
- Provided a clear view of product-level financial performance

Product-Level Analysis

- Evaluated products based on Gross Profit and Gross Margin
- Identified high-performing, low-margin, and underperforming products

Division-Level Analysis

- Compared revenue and profit contribution across divisions
- Highlighted imbalances and underperforming categories

Pareto Analysis

- Identified products contributing to ~80% of revenue and profit
- Revealed strong dependency on a limited number of products

Cost Structure Diagnostics

- Analyzed Cost vs Sales relationships
- Detected cost-heavy, margin-poor products and pricing inefficiencies

Visualization & Dashboard

Insights supported through charts and interactive Streamlit dashboard

Enabled dynamic exploration and data-driven decision-making

Key Insights

Product Performance

- Wonka Bar – Scrumdiddlyumptious is the highest selling product with a 69.44% gross margin and contributes 20.72% of the company's total profit.
- Everlasting Gobstopper shows the highest gross margin but generates very low sales ($130), indicating strong profitability but weak market demand.

Division Performance

- The Chocolate division dominates profitability, contributing 95.06% of total company profit.
- The Sugar category generates the lowest revenue ($427.48) despite maintaining a healthy 66.61% gross margin.

Profit Concentration

- The top five products generate 92% of total sales, indicating a strong dependency on a limited number of products.
- Key factories Lot's O' Nuts and Wicked Choccy contribute 94% of sales but operate at 32.55% margin vs. 40–45% target — urgent cost optimization needed.

Margin Risk

- Kazookles shows a high margin risk with only 7.69% gross margin and an average profit per unit of $0.25.
- Urgent renegotiation for bulk purchasing with Lot's O' Nuts and Wicked Choccy is required to maintain balanced margins and prevent excessive pricing control by these factories.

Recommendations

- Increase production and marketing focus on high-margin products such as Wonka Bar – Scrumdiddlyumptious.
- Investigate demand issues for high-margin but low-volume products like Everlasting Gobstopper.
- Review pricing strategies for low-margin products such as Kazookles.
- Diversify revenue sources to reduce dependency on top-performing products.
- Conduct supplier renegotiation for cost-heavy products.
- Over-reliance on two factories (Lot's O' Nuts and Wicked Choccy) introduces supply-side risk and reduces bargaining power, potentially leading to unfavorable pricing or the need for aggressive cost negotiations in bulk procurement.

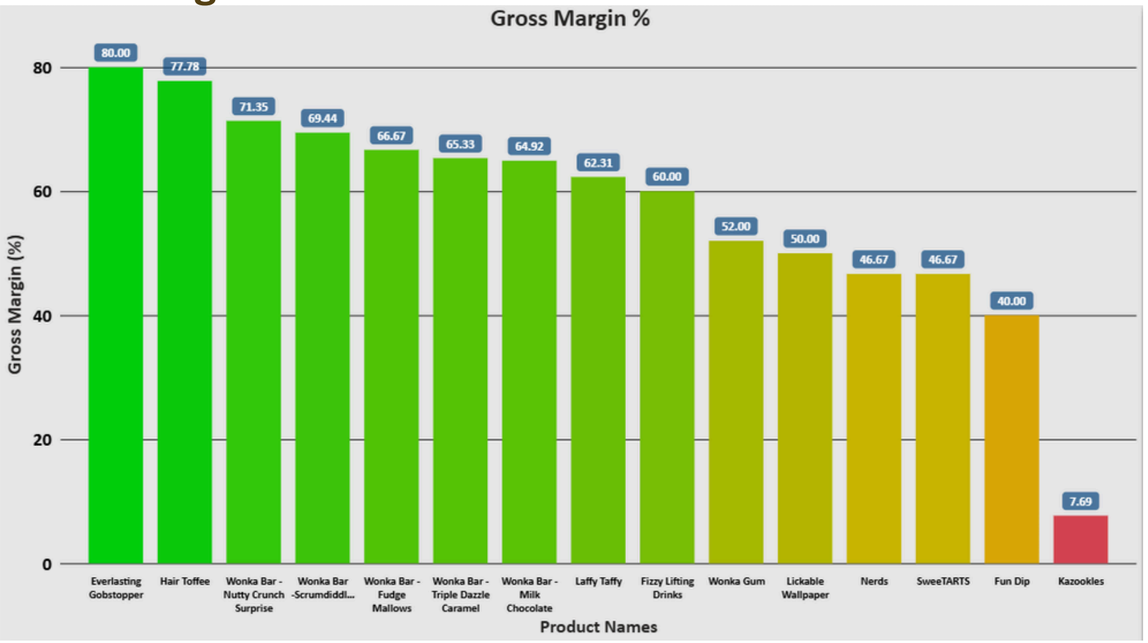
Profit Concentration (Pareto) Analysis

No	Product Name	Sales	Cumulative Sales	Cumulative Sales %
1	Wonka Bar - Triple Dazzle Caramel	28485	28485	20.09
2	Wonka Bar - Scrumdiddlyumptious	27874.8	56359.8	39.75
3	Wonka Bar - Milk Chocolate	26867.75	83227.55	58.7
4	Wonka Bar - Fudge Mallows	24890.4	108117.95	76.26

Observations

- Sales are highly concentrated, with 76.26% coming from the top 4 products and over 92% from the top 5. This over-reliance on a narrow product range increases business risk, particularly in scenarios of supplier disruption, price fluctuations, or demand shifts. Such concentration may affect revenue stability and operational resilience.
- Profit contribution is also highly concentrated, with the top 4 products accounting for approximately 77.3% of total profit. This indicates a critical dependency on a small set of high-performing products. Any disruption in the supply, pricing, or demand of these products could severely impact overall profitability and pose a significant business risk.

Gross Margin % Chart



Overall Margin Performance

- Most products maintain a healthy gross margin of 40% or above, indicating strong baseline profitability across the portfolio.
- Kazookles is a clear outlier with a gross margin of only 7.69%, significantly underperforming compared to all other products.

Top Performing Product

- Everlasting Gobstopper has the highest gross margin at 80%, making it the most profitable product in the portfolio.

Alignment of Profitability & Sales

- 3 out of the top 5 highest gross margin products are also part of the top 80% revenue contributors, indicating a strong overlap between high profitability and high demand

.

Mid-Range Margin Opportunity (40–50%)

- Products in the 40%–50% gross margin range ("yellow zone") represent a key improvement opportunity.
- These products are already stable but have potential to move into higher profitability tiers.

Strategic Recommendations

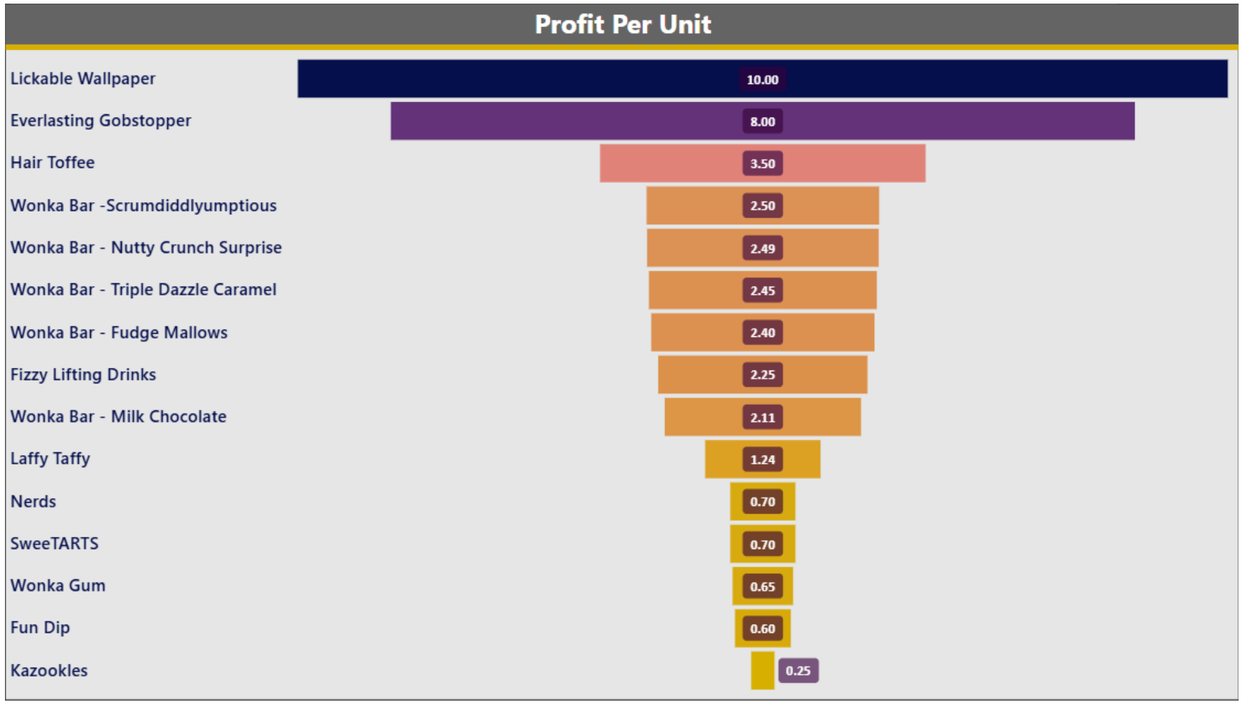
Supplier Negotiation:

- Focus on cost optimization for mid-range products to increase margins from 40–50% → 55–60%.

Marketing Investment:

- Allocate additional marketing budget to these products to boost demand and sales volume.

Profit Per Unit Chart



Key Observations

High Profit per Unit Opportunity

- Lickable Wallpaper has the highest profit per unit at $10 with a gross margin of 50%.
- This indicates a strong premium product position.
- Increasing marketing efforts and demand generation for this product can significantly boost overall revenue and profitability.

High Volume, Low Margin Strategy

- Wonka Bar products are the highest-selling items in the portfolio.
- They generate a profit per unit of $2–$2.5, suggesting a low-margin, high-volume strategy.
- Consumer preference is likely driven by affordability and accessibility, making them key for market penetration and steady revenue flow.

Underperforming Product

- Kazookles has the lowest profit per unit at $0.25, making it the weakest performer in the category.
- This product requires immediate attention for:
  - Cost optimization
  - Pricing strategy revision
  - Or potential discontinuation if profitability cannot be improved

Mid-Tier Profit Optimization (0.5–1.5 Range)

- Products with profit per unit between $0.5–$1.5 represent a key improvement segment.

These products have the potential to :

- Increase profitability through cost control or price adjustments
- Grow through targeted marketing to boost demand

Strategic Recommendations

Demand Generation :

- Invest in marketing for high-margin products like Lickable Wallpaper to maximize returns.

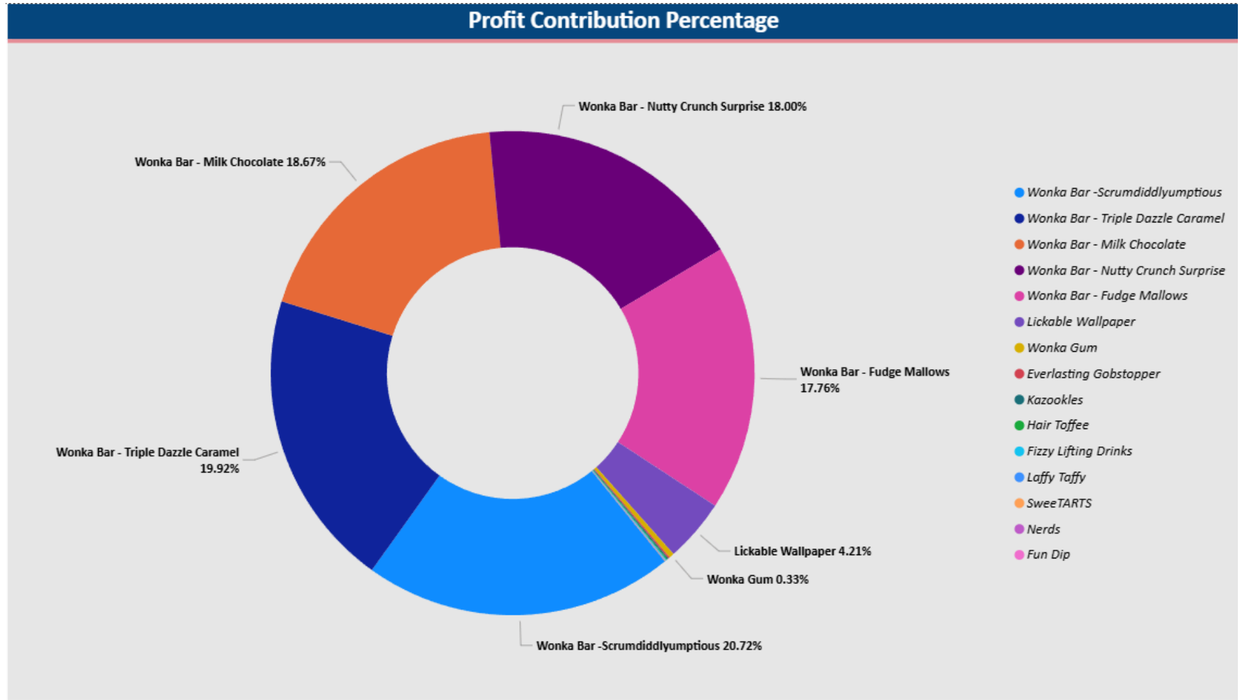
Volume Strategy :

- Continue leveraging Wonka Bar products for high sales volume, while exploring slight margin improvements.

Portfolio Review :

- Deep dive into low and mid-tier products to identify opportunities for profitability enhancement or repositioning.

Total Profit Contribution



Key Observations

High Profit Concentration Risk

- Approximately 92% of the company's total profit is driven by just 5 products, indicating a high dependency on a limited set of products.
- This concentration increases business risk and highlights the need to diversify profit contributors.

Brand-Level Overdependence

- Wonka Bar products dominate the portfolio, contributing over 90% of both revenue and profit.
- This is a significant red flag, as over-reliance on a single brand can:
  - Increase vulnerability to market shifts
  - Reduce negotiation power with suppliers
  - Create brand imbalance within the portfolio

Underperforming Product

- Kazookles contributes less than 0.5% to total company profit, making it a negligible performer.
- This product requires:
  - Performance review
  - Cost and pricing analysis
  - Or potential discontinuation if it remains unviable

Lack of Portfolio Balance

- The current portfolio shows weak contribution from non-Wonka products, indicating missed opportunities in other product lines.
- A more balanced mix of brands/products is required to ensure sustainable growth and reduced risk exposure.

Performance Improvement :

- Invest in marketing, pricing strategy, and positioning of underperforming or mid-tier products.

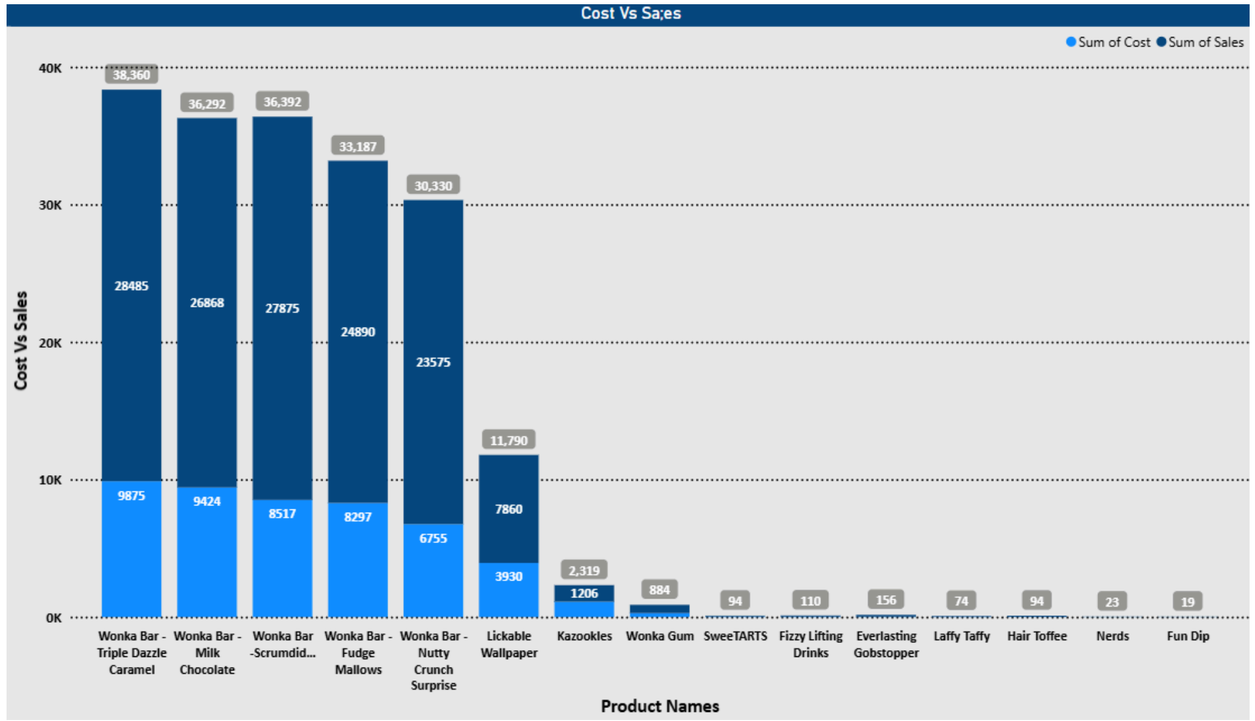
Supplier Risk Management :

- Reducing reliance on a single dominant product/brand will help maintain better negotiation power and financial control.

Portfolio Optimization :

- Identify and scale emerging high-potential products to build a more resilient and balanced portfolio.

Cost Vs Sales Scatter Plot



Key Observations

Strong Cost-to-Sales Efficiency

- All best-selling products demonstrate a healthy cost-to-sales ratio, indicating efficient cost management and strong profitability.

High Return on Low Investment

- The cost of top-selling products is below $10K, while each generates sales exceeding $30K.
- This reflects a high return on investment (ROI) and effective product positioning in the market.

Brand-Level Efficiency

- Wonka Bar products maintain a consistently strong cost-to-sales ratio, reinforcing their role as a financially efficient and reliable brand.

Sales Concentration in a Single Brand

- The top 5 highest-selling products all belong to the Wonka Bar brand.
- This further highlights heavy reliance on a single brand for revenue generation, which may pose a strategic risk if not diversified.

Cost Structure Diagnostics

Cost-heavy, margin-poor products

- Kazookles has a significantly low gross margin of just 7.69%, with a cost-to-sales ratio of 92.3%, making it a highly unprofitable product.
- This indicates that the majority of revenue generated is consumed by costs, leaving minimal contribution to overall profitability

Pricing inefficiencies

Flag products needing:

- Kazookles : Low margin + low demand → optimize or consider discontinuation.

Cost renegotiation

- Kazookles: 7.69% margin and 92.3% cost ratio → highly unprofitable product. Immediate cost optimization is required to improve profitability.

Discontinuation review

- Kazookles: Low margin + low demand → optimize or consider discontinuation.

Congestion-prone states or region

These are regions contributing a large share of total sales—if something goes wrong here, overall performance is heavily impacted.

1. **Pacific (32.66%)**
2. **Atlantic (29.06%)**

Together they contribute ~61.7% of total sales, which means:

- Supply chain delays, stockouts, or demand drops in these regions will create congestion
- Logistics, warehousing, and inventory planning should be strongest here

Over-Dependency Risk

Risk Breakdown :

1. **Top 2 regions (Pacific + Atlantic): 61.7% sales**
2. **Top 3 regions: 84.3% sales**

This indicates business is highly concentrated and Interior + Gulf combined only contribute 38%. If Pacific or Atlantic underperform → major revenue impact.

Conclusion

Over-Dependency Risk

- The company shows significant over-dependency across multiple dimensions, including :
  - Product/brand (heavy reliance on Wonka Bar)
  - Profit contributors (limited number of high-performing products)
  - Suppliers
  - Regional performance (if applicable)

- Such concentration creates a high-risk exposure, where disruption in any one area can significantly impact overall business performance.

- **Supplier Risk & Bargaining Power**
  - Heavy reliance on a limited number of suppliers can reduce the company's bargaining power, especially during periods of high demand.
  - It is essential to :
    - Identify and onboard alternative vendors at competitive pricing
    - Ensure supply continuity and risk mitigation

- This will help the company stay in a safer and more stable operating position.

Margin Improvement Opportunity

- Products in the 40%–50% gross margin range present a strong opportunity for improvement.
- The company should :
  - Focus on cost optimization and supplier negotiations
  - Increase marketing efforts to boost demand, especially for non-seasonal products
- Target should be to move these products into the 55%–60% margin range.

Portfolio Diversification

- The company needs to diversify its product portfolio to reduce reliance on a few high-performing products.
- A balanced mix of products will help :
  - Improve risk management
  - Ensure stable revenue and profit during market fluctuations

Underperforming Product Action

- Kazookles is the only product with:
  - Less than 10% profit margin
  - Lowest sales performance
- It is recommended to :
  - Discontinue the product, or
  - Re-evaluate only if there is a clear turnaround strategy

Customer Data & Analytics System (New Recommendation)

- The company should invest in building a robust data collection and analytics system to better understand customer behavior.
- Key data points to capture:
  - Demographics: Age, gender
  - Behavioral data: Purchase patterns, frequency, preferences
  - Marketing data: Ad performance, campaign engagement, conversion rates
- Benefits:
  - Enable targeted marketing campaigns
  - Improve product positioning and personalization
  - Support data-driven decision-making
  - Identify high-value customer segments

Thank You